A Strict Surface-Energy Obstruction for Timelike Kantowski–Sachs–Schwarzschild Junctions

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Abstract

We prove a strict surface-energy obstruction for timelike spherical junctions between a homogeneous Kantowski–Sachs (KS) region and the ordinary asymptotically flat Schwarzschild exterior. Because the KS areal radius has no longitudinal spatial gradient, $2m_{\mathrm{MS},C}/B \geq 1$ throughout the region. For every matched radius with $F(R) > 0$, every finite shell rapidity, and either child orientation, the angular Israel equation therefore requires $\sigma < 0$. Weak and dominant energy conditions fail independently of surface pressure, without a critical radius or cosmological-constant scale. Null-energy-condition violation remains trajectory dependent. For an exact static comoving family, it occurs precisely outside the Schwarzschild photon sphere, $R_0 > 3m$; a Nariai $dS_2 \times S^2$ realization exists with $\Lambda = R_0^{-2}$, a scale specific to that auxiliary family. The result is a strict KS specialization of established spherical-shell theory and explains why a direct timelike completion to the retained ordinary exterior requires exotic surface energy. It does not apply inside the horizon or to spacelike or wormhole-connected transitions.

Keywords: junction conditions; thin shells; Kantowski–Sachs spacetime; Schwarzschild spacetime; surface energy conditions

1. Introduction

Matching spacetime regions across a hypersurface is a standard construction in gravitational physics. When the induced metric is continuous but the extrinsic curvature jumps, the jump is supported by a distributional surface stress tensor governed in general relativity by the Israel conditions [1]. The method is used for stellar boundaries, collapsing shells, vacuum bubbles, wormholes, regular black holes, and proposed black-hole-to-cosmology transitions.

The last application requires careful causal and global bookkeeping. A transition surface may be timelike, spacelike, or null; the retained side of each geometry fixes the physically relevant normal orientations; and expansion of total spatial volume need not imply expansion of the areal radius. A Kantowski–Sachs representation does not by itself establish an expanding cosmology, and a Schwarzschild-time divergence at a horizon does not establish a reversed time direction.

Berezin, Kuzmin, and Tkachev (BKT) [2] derive invariant outer-curvature and junction equations valid for arbitrary spherically symmetric shells, including dynamical geometries and changes between ordinary and wormhole branches. Farhi and Guth [3], Blau, Guendelman, and Guth [4], and Farhi, Guth, and Guven [5] establish the closest physical lineage: the classical obstruction to producing an inflating universe on the ordinary asymptotic branch, the global dynamics of false-vacuum bubbles, and quantum access to the expanding wormhole-connected branch. Rosa and Carloni [6] give a complementary covariant treatment for locally rotationally symmetric spacetimes and boundaries of all causal types. The present paper does not replace or generalize those frameworks. It proves their strict consequence for a homogeneous Kantowski–Sachs side joined to the retained asymptotically flat Schwarzschild exterior. Related work has used Kantowski–Sachs variables for distinct null-shell problems [7]. Historic black-hole-to-cosmology constructions instead place the transition on a spacelike surface inside the black hole [8,9].

The same distinction is relevant to quantum-corrected black-hole interiors. Representative loop-quantum-gravity models treat the Schwarzschild interior as a homogeneous KS system: the classical singularity is replaced by a Nariai-type late-time region [14], a bounce into a white-hole interior [15], or a transition surface separating trapped and anti-trapped regions within a larger effective extension [16]. An explicit Schwarzschild-to-KS construction using a localized layer instead places a spacelike S-brane inside the horizon [17]. None of these models uses the timelike ordinary-$F > 0$ junction studied here. The present theorem does not test their quantum dynamics; it supplies the classical boundary diagnostic for a different completion in which a KS region would be attached directly to the retained ordinary exterior.

We show why the corresponding direct ordinary-exterior timelike construction is obstructed. BKT's invariant formula already contains the underlying signed-root mechanism [2, p. 2925, Eq. (2.54a)]. In the Kantowski–Sachs specialization, the absence of a longitudinal areal-radius gradient makes the comparison strict: the magnitude of the child angular extrinsic curvature is smaller than the Schwarzschild exterior contribution whenever $F > 0$. Retaining the asymptotically flat exterior fixes the parent-normal direction, so the angular jump is positive for either child-side orientation. Negative surface energy density is therefore universal within the theorem domain; NEC violation is a separate, stronger conclusion that requires additional trajectory data.

The principal result is the KS sign theorem and its implication for the direct ordinary-exterior construction. The derivation fixes the exterior orientation from the retained global region, obtains both independent mixed curvature components, closes the Israel and Codazzi relations, contrasts the KS product structure with FRW geometry, treats turning points without division by $\dot{R}$, and reproduces the exterior curvature in ingoing Eddington–Finkelstein coordinates.

The theorem applies to the comoving interface and to any finite radial shell trajectory satisfying the declared embedding and areal matching relation, while $R > 0$ and $F > 0$. It does not cover $F < 0$, spacelike transitions, alternative global gluings, finite-thickness layers, or independent surface-spin sectors.

One interpretive warning will matter later: Kantowski–Sachs total-volume expansion, governed by $\Theta = H_A + 2H_B$, does not imply expansion of the symmetry spheres, which requires $H_B > 0$. Section 9 gives the corresponding shear identity and explains why this distinction prevents the junction calculation from being read as evidence for an areal bounce.

2. Geometry and conventions

We use signature $(-+++)$. The Kantowski–Sachs metric is

$$ds_C^2 = -N^2 dt_C^2 + A^2(t_C)d\chi^2 + B^2(t_C)d\Omega_2^2. \tag{1}$$

After deriving the normalization condition we choose child proper time $N = 1$, with

$$H_A = \frac{1}{A}\frac{dA}{dt_C}, \qquad H_B = \frac{1}{B}\frac{dB}{dt_C}, \qquad \Theta = H_A + 2H_B. \tag{2}$$

The moving child embedding is

$$X_C^\mu = (t_C(\tau), \chi(\tau), \theta, \phi), \qquad R(\tau) = B(t_C(\tau)). \tag{3}$$

The comoving case has $\dot{\chi} = 0$; the moving case permits an otherwise arbitrary finite radial trajectory subject to Eq. (3). A dot denotes $d/d\tau$, where τ is shell proper time. Both embeddings are future oriented: we choose $\dot{t}_C > 0$, and in the $F > 0$ Schwarzschild region we choose $\dot{T} > 0$. Thus Eq. (15) uses the positive root $\gamma = \dot{t}_C = \sqrt{1+X^2}$, while Eq. (10) uses $\dot{T} = \beta/F > 0$.

The Schwarzschild parent is

$$ds_P^2 = -F(R)dT^2 + \frac{dR^2}{F(R)} + R^2 d\Omega_2^2, \qquad F(R) = 1 - \frac{2m}{R}, \tag{4}$$

where $m = GM/c^2$. Its shell embedding is

$$X_P^\mu = (T(\tau), R(\tau), \theta, \phi). \tag{5}$$

The theorem concerns $F(R) > 0$. The limit $F \to 0^+$ is treated only as a chart-control limit.

The first junction condition is imposed explicitly. Pulling back either bulk metric to the shell gives the common induced metric $ds_\Sigma^2 = -d\tau^2 + R^2(\tau)d\Omega_2^2$; equality of the angular components is precisely the areal matching relation $R(\tau) = B(t_C(\tau))$ in Eq. (3). The temporal component is the proper-time normalization used below on each side.

The surface tensor is

$$\Sigma^a{}_b = \mathrm{diag}(-\sigma, p_s, p_s). \tag{6}$$

We define

$$[K^a{}_b] = K^a{}_{b,P} - K^a{}_{b,C} \tag{7}$$

and use

$$[K_{ab}] - h_{ab}[K] = -\frac{8\pi G}{c^4}\Sigma_{ab}, \qquad K_{ab} = e_a{}^\mu e_b{}^\nu \nabla_\mu n_\nu. \tag{8}$$

The convention is anchored by $K^\theta{}_\theta = +1/R$ for the outward normal to a round sphere in flat space. Orientation signs are denoted $\epsilon_P, \epsilon_C \in \{-1, +1\}$.

Table 1: Core conventions.

Object	Convention
Metric signature	$(-+++)$
Jump	$[K^a{}_b] = K^a{}_{b,P} - K^a{}_{b,C}$
Surface tensor	$\Sigma^a{}_b = \mathrm{diag}(-\sigma, p_s, p_s)$
Extrinsic curvature	$K_{ab} = e_a{}^\mu e_b{}^\nu \nabla_\mu n_\nu$
Ordinary retained exterior	$\epsilon_P = +1$
Theorem domain	$F(R) > 0$, finite real X

Figure 1 summarizes the declared geometry, retained regions, and normal orientations.

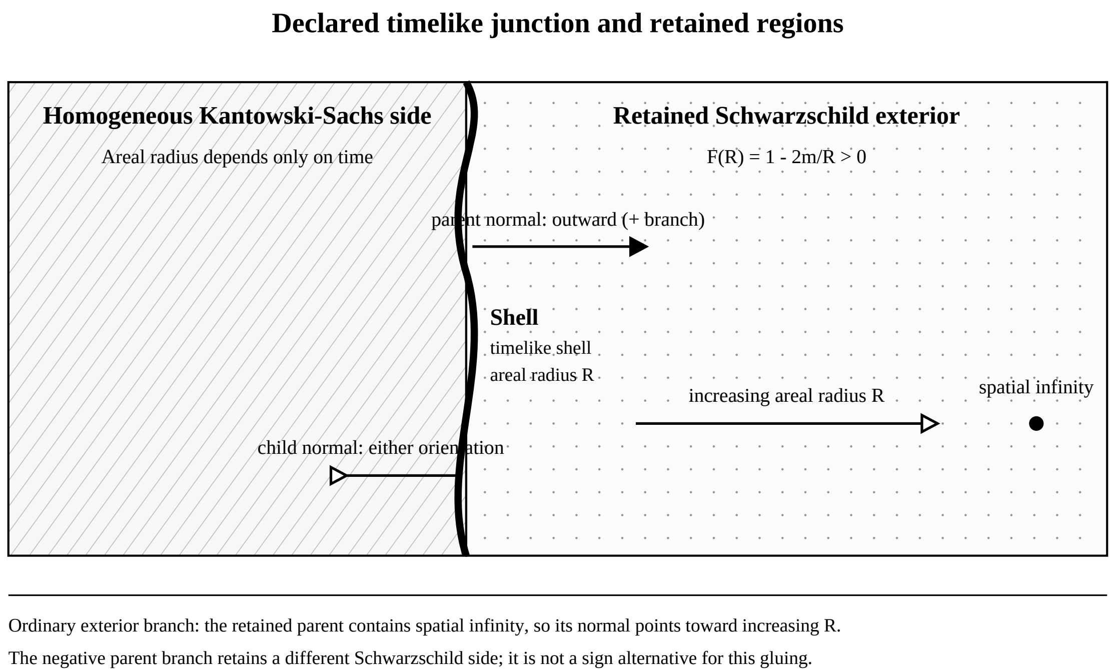

Figure 1: Declared spherical timelike junction. The retained Schwarzschild parent contains spatial infinity, fixing its normal toward increasing areal radius and hence $\epsilon_P = +1$. The $\epsilon_P = -1$ algebraic branch retains a different Schwarzschild side; it is not a sign alternative for the same global gluing. The child-side orientation remains explicit through $\epsilon_C = \pm 1$.

3. Exterior-orientation lemma

Lemma 1. If the retained parent region is the Schwarzschild exterior containing spatial infinity and the normal points from the shell into that retained region, then

$$\epsilon_P = +1. \tag{9}$$

Proof. The areal coordinate increases from the shell toward the asymptotically flat end. The parent normal derived below has $n_P^R = \epsilon_P \beta$, with $\beta > 0$. Hence a normal directed into the retained exterior requires $\epsilon_P = +1$. □

The algebraic branch $\epsilon_P = -1$ retains the opposite side or describes a throat/back-to-back gluing; it is the branch used for wormhole-connected false-vacuum-bubble constructions [4,5]. It is not an alternative convention for the same exterior. Simultaneously reversing both normals and the jump order leaves the physical surface tensor unchanged.

The child sign has a different status. Once a KS region and a timelike boundary are specified, ϵ_C records which of the two local KS sides is retained and hence which unit normal points from the shell into that retained child region. The alternatives $\epsilon_C = \pm 1$ therefore describe distinct retained-side embeddings, not an arbitrary reversal of one normal while holding the gluing fixed. In both cases the jump order remains parent minus child as declared in Eq. (7), and the parent normal remains fixed by the retained asymptotically flat exterior. Theorem 1 proves the same sign conclusion for either admissible child-side choice.

4. Complete extrinsic curvature

4.1 Schwarzschild side

Proper-time normalization gives

$$F\dot{T}^2 - \frac{\dot{R}^2}{F} = 1. \tag{10}$$

Define

$$\beta = \sqrt{F + \dot{R}^2}, \qquad \dot{T} = \frac{\beta}{F}. \tag{11}$$

An oriented unit normal covector is

$$n_{\mu,P} = \epsilon_P \left(-\dot{R}, \frac{\beta}{F}, 0, 0 \right), \tag{12}$$

with $n_P^R = \epsilon_P \beta$. Direct calculation gives

$$\boxed{K^\tau{}_{\tau,P} = \epsilon_P \frac{\ddot{R} + m/R^2}{\beta}, \qquad K^\theta{}_{\theta,P} = \epsilon_P \frac{\beta}{R}.} \tag{13}$$

The acceleration form of $K^\tau{}_{\tau,P}$ supplies its regular turning-point value without dividing by $\dot{R}$.

4.2 Moving Kantowski–Sachs side

Child-side normalization is

$$\dot{t}_C^2 - A^2\dot{\chi}^2 = 1. \tag{14}$$

Introduce

$$X = A\dot{\chi}, \qquad \gamma = \dot{t}_C = \sqrt{1 + X^2}. \tag{15}$$

The oriented unit normal is

$$n_{\mu,C} = \epsilon_C(-A\dot{\chi}, A\dot{t}_C, 0, 0). \tag{16}$$

Coordinate evaluation gives

$$\boxed{K^\tau{}_{\tau,C} = \epsilon_C \left(\frac{\dot{X}}{\gamma} + H_A X \right), \qquad K^\theta{}_{\theta,C} = \epsilon_C X H_B.} \tag{17}$$

Differentiating the areal matching condition yields

$$\frac{\dot{R}}{R} = \gamma H_B, \tag{18}$$

so

$$\boxed{K^{\theta}{}_{\theta,C} = \epsilon_C \frac{X\dot{R}}{\gamma R}.} \tag{19}$$

An independent orthonormal-frame derivation uses

$$u^{\hat{a}} = (\gamma, X), \qquad n^{\hat{a}} = \epsilon_C(X, \gamma) \tag{20}$$

and reproduces both components exactly. In the comoving limit $X = \dot{X} = 0$, both child components vanish.

Table 2: Mixed extrinsic-curvature components.

Side	$K^{\tau}{}_{\tau}$	$K^{\theta}{}_{\theta}$
Schwarzschild parent	$\epsilon_P(\ddot{R} + m/R^2)/\beta$	$\epsilon_P \beta/R$
Moving KS child	$\epsilon_C(\dot{X}/\gamma + H_A X)$	$\epsilon_C X H_B = \epsilon_C X\dot{R}/(\gamma R)$
Comoving KS child	0	0

5. Thin-shell obstruction

The invariant origin of the angular comparison is BKT's general formula $K^{\theta}{}_{\theta} = R^{-1} n^{\mu} \nabla_{\mu} R$ and their angular Israel equation [2, p. 2925, Eqs. (2.54a) and (2.59a)]. BKT's orientation sign records whether areal radius increases or decreases along the outward normal, the same geometric distinction encoded here by ϵ_P and ϵ_C; our jump is ordered parent minus child as in Eq. (7), and Eq. (8) fixes the corresponding Israel sign. The following theorem is the strict Kantowski–Sachs–Schwarzschild specialization.

Theorem 1 (KS ordinary-exterior specialization). Consider the moving embedding (3) of a homogeneous Kantowski–Sachs region into a Schwarzschild exterior with $R > 0$ and $F(R) > 0$. Retain the parent region containing spatial infinity and use conventions (6)–(8). For every finite real X,

$$\Delta K_{\theta} \equiv K^{\theta}{}_{\theta,P} - K^{\theta}{}_{\theta,C} > 0 \tag{21}$$

for either child-side orientation. Consequently $\sigma < 0$ for every shell satisfying these hypotheses, independently of surface pressure and shell acceleration. This universal sign conclusion does not by itself imply $\sigma + p_s < 0$; NEC violation requires the additional condition $\Delta K_{\tau} - \Delta K_{\theta} < 0$.

Proof. Since $\gamma = \sqrt{1 + X^2}$,

$$\frac{|X|}{\gamma} < 1. \tag{22}$$

Equation (19) therefore gives

$$\left| K^{\theta}{}_{\theta,C} \right| < \frac{|\dot{R}|}{R}. \tag{23}$$

Because $F > 0$,

$$\frac{|\dot{R}|}{R} < \frac{\sqrt{F + \dot{R}^2}}{R} = \frac{\beta}{R}. \tag{24}$$

Lemma 1 fixes $K^{\theta}{}_{\theta,P} = \beta/R > 0$. Hence

$$\Delta K_{\theta} \geq \frac{\beta}{R} - \left| K^{\theta}{}_{\theta,C} \right| > 0 \tag{25}$$

for either ϵ_C. The Israel equation then gives

$$\boxed{\sigma = -\frac{c^4}{4\pi G} \Delta K_{\theta} < 0.} \tag{26}$$

□

Corollary 1 (comoving interface). In the comoving limit $X = 0$, the child angular curvature vanishes and

$$\Delta K_{\theta} = \frac{\beta}{R} > 0, \qquad \sigma < 0. \tag{27}$$

Thus the ordinary-exterior obstruction includes the comoving interface as a strict special case.

5.1 Invariant mass interpretation

The direct proof above is decisive because it fixes the signed Israel jump and surface tensor. The same geometry admits a useful invariant cross-check through the Misner–Sharp geometrized mass length $m_{\mathrm{MS}} = GM_{\mathrm{MS}}/c^2$, defined in spherical symmetry by

$$1 - \frac{2m_{\mathrm{MS}}}{R} = g^{\mu\nu} \nabla_{\mu} R \nabla_{\nu} R. \tag{28}$$

For any spherical timelike shell, decomposition of the areal-radius gradient into tangential and normal parts gives

$$g^{\mu\nu} \nabla_{\mu} R \nabla_{\nu} R = -\dot{R}^2 + \left(R K^{\theta}{}_{\theta} \right)^2. \tag{29}$$

On the Schwarzschild side, Eq. (28) gives $m_{\mathrm{MS},P} = m$, and $F > 0$ is equivalent to $2m/R < 1$. On the homogeneous Kantowski–Sachs side, $R = B(t_C)$ has no longitudinal spatial gradient. Consequently, throughout the KS region,

$$g_C^{\mu\nu} \nabla_{\mu} R \nabla_{\nu} R = -\left(\frac{dB}{dt_C} \right)^2, \tag{30}$$

and therefore

$$\frac{2m_{\mathrm{MS},C}}{B} = 1 + \left(\frac{dB}{dt_C} \right)^2 \geq 1, \tag{31}$$

with equality only where $dB/dt_C = 0$. This is a property of the KS region, independent of any shell embedding. On the shell, the areal matching relation gives

$$\left(\frac{dB}{dt_C}\right)^2 = \frac{\dot{R}^2}{\gamma^2}. \tag{32}$$

Thus the KS symmetry spheres are marginal, future trapped, or past trapped (anti-trapped), depending on the time orientation and the sign of dB/dt_C. The retained $F > 0$ Schwarzschild exterior is untrapped. Their mass difference at the common areal radius is

$$\frac{2(m_{\mathrm{MS},C} - m_{\mathrm{MS},P})}{R} = F + \frac{\dot{R}^2}{\gamma^2} > 0. \tag{33}$$

Equations (28)–(33) supply an invariant physical reading of the signed-curvature proof: the shell is asked to join the KS non-untrapped sector, with greater Misner–Sharp mass length at the matched radius, to an ordinary untrapped exterior with smaller enclosed Schwarzschild mass length. When combined with the declared orientation and signed Israel equation, this mismatch corresponds to negative σ. The mass ordering alone does not determine the orientation sign of $K^\theta{}_\theta$, which is why Theorem 1 remains the primary proof. This is also the point of contact with BKT's general invariant formula [2]: our result is its strict KS/ordinary-exterior consequence, not an independent generalization. The complementary $F < 0$ Schwarzschild region is trapped or anti-trapped, with the classification fixed by time orientation.

5.2 Why homogeneity does not impose the same result in FRW

The obstruction is not a consequence of spatial homogeneity alone. In a Friedmann–Robertson–Walker region,

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2 d\Omega_2^2\right), \qquad R = a(t)r, \tag{34}$$

the areal radius has both temporal and radial gradients. Hence

$$g^{\mu\nu}\nabla_\mu R \nabla_\nu R = -H^2 R^2 + 1 - kr^2, \tag{35}$$

and the positive radial term can make the symmetry spheres untrapped. The KS step in Eq. (30), where the longitudinal spatial derivative vanishes identically, therefore has no automatic FRW analogue. BKT's general treatment and FRW appendix already contain the corresponding outer-curvature formulas [2, pp. 2942–2943, Appendix B]. The comparison here identifies the KS geometric property that makes inequality (25) strict; it is not a new FRW junction result.

6. Surface tensor and energy conditions

Define

$$\Delta K_\tau = K^\tau{}_{\tau,P} - K^\tau{}_{\tau,C}. \tag{36}$$

The complete mixed Israel solution is

$$\boxed{\sigma = -\frac{c^4}{4\pi G}\Delta K_\theta, \qquad p_s = \frac{c^4}{8\pi G}\left(\Delta K_\tau + \Delta K_\theta\right).} \tag{37}$$

All independent tensor residuals vanish after substitution. Useful combinations are

$$\sigma + p_s = \frac{c^4}{8\pi G}\left(\Delta K_\tau - \Delta K_\theta\right), \tag{38}$$

$$\sigma + 2p_s = \frac{c^4}{4\pi G}\Delta K_\tau. \tag{39}$$

6.1 Null-energy criterion and the static comoving family

The shell NEC is controlled by the signed scalar

$$\begin{aligned} \mathcal{N} &\equiv \Delta K_\tau - \Delta K_\theta \\ &= \frac{R\ddot{R} - \dot{R}^2 - 1 + 3m/R}{\beta R} - \epsilon_C\left(\frac{\dot{X}}{\gamma} + H_A X - \frac{X\dot{R}}{\gamma R}\right). \end{aligned} \tag{40}$$

Equation (38) shows that the NEC is satisfied when $\mathcal{N} \geq 0$ and violated when $\mathcal{N} < 0$. The logical hierarchy is therefore sharp: $\sigma < 0$ is universal under Theorem 1, whereas NEC violation is not. The sign of $\mathcal{N}$ depends additionally on the shell trajectory, acceleration, and child-side kinematics and orientation.

Proposition 2 (static comoving family). Fix $m > 0$ and $R_0 > 2m$. Suppose the declared junction contains an exact static comoving boundary segment on which

$$X = \dot{X} = 0, \qquad R(\tau) = B(t_C(\tau)) = R_0, \qquad \dot{R} = \ddot{R} = 0.$$

Then the surface tensor is given by Eq. (41b). Its NEC and $2+1$-dimensional SEC are satisfied for $2m < R_0 < 3m$, saturated at $R_0 = 3m$, and violated for $R_0 > 3m$. WEC and DEC fail throughout the family.

Proof. This hypothesis is stronger than an instantaneous turning point: B is constant along an open segment of the comoving boundary. Both child extrinsic-curvature components vanish, whereas the ordinary-exterior Schwarzschild components remain nonzero. Equation (40) gives

$$\mathcal{N}_{\text{static}} = \frac{3m - R_0}{R_0^2\sqrt{F_0}}, \qquad F_0 = 1 - \frac{2m}{R_0}. \tag{41a}$$

The complete surface stress tensor for this family is

$$\sigma_0 = -\frac{c^4}{4\pi G}\frac{\sqrt{F_0}}{R_0}, \qquad p_{s0} = \frac{c^4}{8\pi G}\frac{R_0 - m}{R_0^2\sqrt{F_0}}, \qquad w_s \equiv \frac{p_{s0}}{\sigma_0} = -\frac{R_0 - m}{2(R_0 - 2m)}. \tag{41b}$$

Consequently,

$$\sigma_0 + p_{s0} = \frac{c^4}{8\pi G}\frac{3m - R_0}{R_0^2\sqrt{F_0}}, \qquad \sigma_0 + 2p_{s0} = \frac{c^4}{4\pi G}\frac{m}{R_0^2\sqrt{F_0}} > 0. \tag{41c}$$

Within the theorem domain, the shell NEC is therefore violated for $R_0 > 3m$, saturated at $R_0 = 3m$, and satisfied for $2m < R_0 < 3m$.

Equivalently, $-1 < w_s < -1/2$ in the NEC-violating interval, $w_s = -1$ at the threshold, and $w_s < -1$ below it. These ratios classify the surface tensor algebraically; because $\sigma_0 < 0$, they should not be read as ordinary positive-density fluid equations of state. The second relation in Eq. (41c) shows that the second $2+1$-dimensional SEC inequality is satisfied throughout the family, so its SEC disposition is controlled entirely by the NEC. This proves the proposition. □

The threshold $R_0 = 3m$ is the Schwarzschild photon-sphere radius, so this static family violates NEC precisely outside the photon sphere. Figure 2 displays the four surface-stress combinations across the family. The same algebraic boundary appears in the distinct construction of a static Schwarzschild thin-shell wormhole, for which NEC is satisfied at the throat when $2m < a_0 \leq 3m$ [13, Appendix A]. That construction glues two Schwarzschild exteriors with wormhole orientations; Proposition 2 instead has vanishing child curvature from a KS product region and retains one ordinary exterior. The shared threshold is therefore attributed to the static Schwarzschild curvature combination, not claimed as unique to KS.

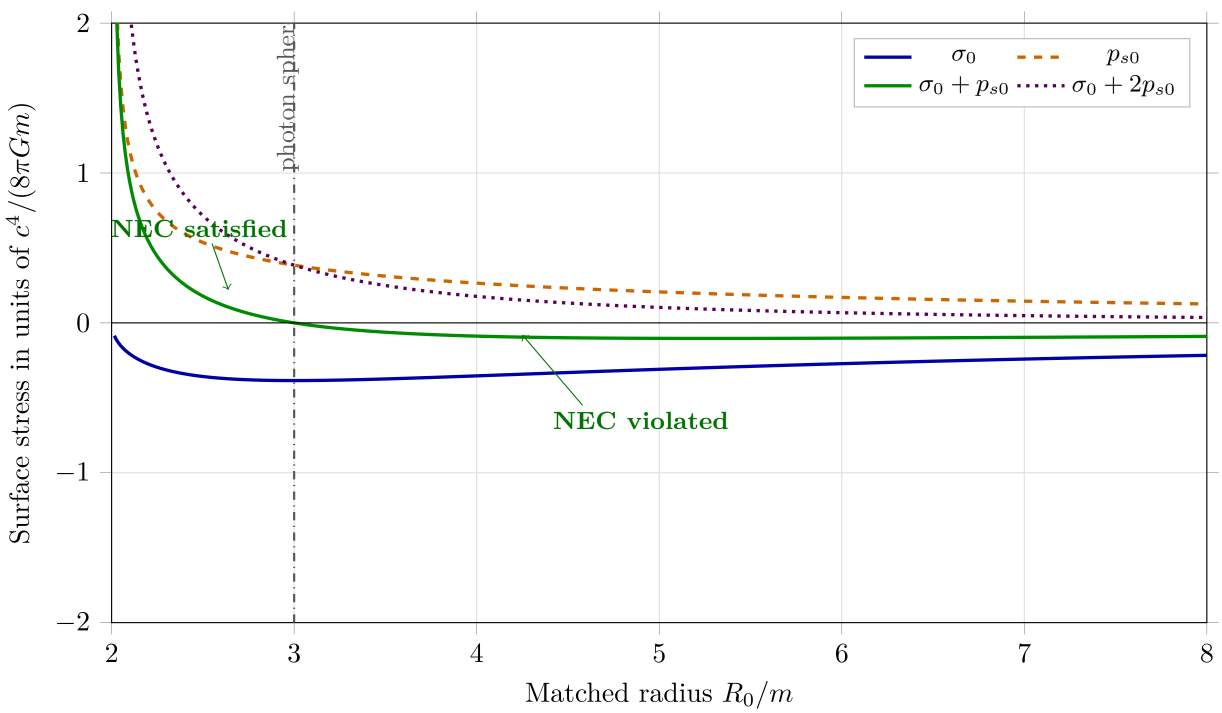

Figure 2: Dimensionless surface stresses for the exact static comoving family as functions of R_0/m. The surface density remains negative and $\sigma_0 + 2p_{s0}$ remains positive throughout $R_0 > 2m$, while $\sigma_0 + p_{s0}$ changes sign at the Schwarzschild photon sphere, $R_0 = 3m$. The plotting window truncates the pressure divergences as $R_0 \to 2m^+$.

The family is not merely a pointwise kinematic assignment. An explicit isotropic bulk realization is obtained by taking $B = R_0$ throughout a KS interval. In $c = 1$ units, the independent Einstein equations then reduce to

$$\kappa\rho = \frac{1}{R_0^2}, \qquad \kappa p_\chi = -\frac{1}{R_0^2}, \qquad \kappa p_\perp = -\frac{\ddot{A}}{A}. \tag{41d}$$

Isotropy is therefore achieved with $p_\chi = p_\perp = -\rho$ provided

$$\frac{\ddot{A}}{A} = \frac{1}{R_0^2}, \qquad A(t_C) = A_+ e^{t_C/R_0} + A_- e^{-t_C/R_0}, \tag{41e}$$

on any interval where $A > 0$. This is the KS form of the Nariai $dS_2 \times S^2$ product region [12], with $\Lambda = R_0^{-2}$. Its comoving boundary has zero child extrinsic curvature and zero matter flux, so Eqs. (41b)–(41c) give the complete Israel layer required to attach it to a Schwarzschild exterior with vanishing cosmological constant. Each value of R_0 fixes a different vacuum-energy scale; this is not a radius family within one theory with fixed Λ. The scale-free statement belongs only to Theorem 1, whereas Proposition 2 is a special product solution with a radius-dependent bulk scale. This example establishes existence of the local bulk geometry and junction data; it does not establish radial stability or promote the $3m$ threshold to generic KS evolution.

At the level of thin-wall architecture, this realization parallels the false-vacuum interior/true-vacuum exterior problem studied by Blau, Guendelman, and Guth [4]. The comparison is structural rather than an identification: their interior is four-dimensional de Sitter geometry and their analysis includes the ordinary and wormhole-connected global branches, whereas Proposition 2 uses a $dS_2 \times S^2$ product interior and isolates the ordinary-exterior static surface tensor. The photon-sphere threshold is therefore a result of this KS static specialization, not a critical radius imported from the de Sitter bubble problem.

More generally, Eqs. (41a)–(41c) give the required surface stress whenever the declared geometry contains an exact static comoving segment, without assuming this particular isotropic realization. They identify a nonempty, orientation-independent sector in which negative surface density is accompanied by $\sigma + p_s < 0$.

For the isotropic 2 + 1-dimensional shell, WEC requires $\sigma \geq 0$ and $\sigma + p_s \geq 0$, while DEC requires $\sigma \geq |p_s|$. Theorem 1 therefore proves failure of both WEC and DEC, regardless of p_s. NEC requires $\mathcal{N} \geq 0$. The shell SEC requires both $\mathcal{N} \geq 0$ and $\Delta K_\tau \geq 0$. Thus NEC and SEC remain trajectory-dependent in general, while Eq. (41a) gives an explicit NEC-violating subfamily.

Table 3: Surface energy-condition disposition on the ordinary exterior branch.

Condition	Requirement	Result from Theorem 1
NEC	$\sigma + p_s \geq 0$	Trajectory-dependent; violated by the static comoving family for $R_0 > 3m$
WEC	$\sigma \geq 0$ and NEC	Violated because $\sigma < 0$
DEC	$\sigma \geq \|p_s\|$	Violated because $\sigma < 0$
SEC in $2 + 1$	NEC and $\sigma + 2p_s \geq 0$	Trajectory-dependent; for the static comoving family it fails exactly when $R_0 > 3m$

7. Surface conservation

The intrinsic shell metric is

$$ds^2_\Sigma = -d\tau^2 + R^2(\tau)d\Omega_2^2. \tag{42}$$

Its surface divergence is

$$D_a\Sigma^a{}_\tau = -\left[\dot\sigma + 2\frac{\dot R}{R}(\sigma + p_s)\right]. \tag{43}$$

The factor of two is the area-expansion factor for the shell's two angular directions. The distributional Bianchi identity requires

$$D_a\Sigma^a{}_b + [T_{\mu\nu}n^\mu e^\nu{}_b]^P_C = 0. \tag{44}$$

The Schwarzschild side is vacuum. For an isotropic perfect fluid comoving with the Kantowski–Sachs coordinates, the child energy and longitudinal-pressure equations are

$$\kappa\rho = 2H_AH_B + H_B^2 + \frac{1}{B^2}, \qquad \kappa p = -2\frac{dH_B}{dt_C} - 3H_B^2 - \frac{1}{B^2}.$$

Their sum gives

$$\frac{dH_B}{dt_C} - H_AH_B + H_B^2 = -\frac{\kappa}{2}(\rho + p). \tag{45}$$

Using Eqs. (17)–(19), the Codazzi identity reduces in $c = 1$ units to

$$\boxed{\dot\sigma + 2\frac{\dot R}{R}(\sigma + p_s) = -\epsilon_C\gamma X(\rho + p).} \tag{46}$$

Direct projection of the perfect-fluid stress gives the same bulk flux. The right-hand side is the kinematic flux measured when a moving shell crosses comoving child matter. It is not a parent-to-child deposition law. In the comoving limit, $X = 0$ and the flux vanishes.

8. Turning point and regular-chart controls

At an exterior turning point, $\dot R = 0$. Equation (18) then implies $H_B = 0$, and

$$K^\theta{}_{\theta,C} = 0, \qquad K^\theta{}_{\theta,P} = \epsilon_P\frac{\sqrt{F}}{R}. \tag{47}$$

The ordinary branch retains $\sigma < 0$. The pressure remains finite for finite $\ddot R, \dot X, H_A$, and X.

To control the Schwarzschild horizon coordinate singularity, introduce

$$v = T + r_*(R), \qquad \frac{dr_*}{dR} = \frac{1}{F}. \tag{48}$$

Then

$$ds_P^2 = -F\,dv^2 + 2\,dv\,dR + R^2d\Omega_2^2. \tag{49}$$

Proper-time normalization gives

$$-F\dot{v}^2 + 2\dot{v}\dot{R} = -1, \tag{50}$$

with future-ingoing root

$$\dot{v} = \frac{\beta + \dot{R}}{F} = \frac{1}{\beta - \dot{R}}. \tag{51}$$

Using

$$n_{\mu,P} = \epsilon_P(-\dot{R}, \dot{v}, 0, 0), \tag{52}$$

direct calculation reproduces Eq. (13) exactly. For a genuinely infalling timelike shell with nonzero $\dot{R} < 0$, $\beta \to -\dot{R}$ as $F \to 0^+$, so

$$\dot{v} \longrightarrow -\frac{1}{2\dot{R}}, \tag{53}$$

which is finite. The surface tensor also remains finite for finite acceleration. This is a regular-chart limit, not an extension of Theorem 1 into $F < 0$, and it implies no reversed time direction.

9. Relation to prior work and limitations

The affirmative KS result is geometric and region-wide. Because the Kantowski–Sachs product geometry has areal radius $B(t_C)$ with no longitudinal spatial gradient, Eq. (31) shows that $2m_{\mathrm{MS},C}/B \geq 1$ identically throughout the region, with equality only where $dB/dt_C = 0$. The shell relation $R = B$ is imposed only when evaluating a matched event. The obstruction therefore applies at every matched radius in the ordinary $F > 0$ exterior, with no critical radius or cosmological-constant scale.

BKT [2] supply the general invariant angular-curvature formula and its surface-energy junction equation for arbitrary spherical shells; consequently neither the signed-root mechanism nor its availability for dynamical spherical interiors is claimed as new here. Rosa and Carloni [6] supply the encompassing covariant LRS framework. The result proved here is the explicit comoving/moving KS specialization and the strict sign inequality (25) for the ordinary retained Schwarzschild exterior. The FRW comparison identifies the product-geometric premise, while the orientation, conservation, turning-point, and regular-chart calculations close the physical statement. We have not found a prior publication stating the same KS/ordinary-exterior theorem. The distinct Schwarzschild thin-shell-wormhole result [13] already contains the static $3m$ NEC threshold; Proposition 2 therefore claims the KS/Nariai surface tensor and its interpretation, not novelty for that radius alone.

The $F > 0$ restriction is the invariant untrapped-parent condition, not an arbitrary cutoff. For $F < 0$, the standard Schwarzschild black-hole time orientation is future trapped and the time-reversed white-hole orientation is past trapped. The strict comparison between the KS non-untrapped sector and the untrapped Schwarzschild exterior used in Eqs. (24) and (33) therefore lifts in the complementary problem. The spacelike layers used by Frolov, Markov, and Mukhanov [8,9] pose a different junction problem inside that regime. The theorem does not obstruct their chosen architecture; it explains why the corresponding ordinary-exterior timelike alternative needs exotic support.

The $\epsilon_P = -1$ algebraic branch corresponds, after translating retained-region and normal conventions, to the wormhole-connected global constructions studied by Blau, Guendelman, and Guth and by Farhi, Guth, and Guven [4,5]. It is the branch on which an expanding child region can be attached without retaining the same ordinary asymptotically flat side used in Theorem 1. It must therefore be supplied with its own global regions and physical interpretation rather than treated as a sign alternative for the present exterior.

The Farhi–Guth past-incompleteness argument concerns the global causal structure and bulk energy conditions of inflating spacetimes [3]. Determining its applicability to the Kantowski–Sachs interiors considered here is outside the present thin-shell analysis, which establishes only the sign of the distributional surface energy for the declared timelike matching.

Finally, negative σ is a distributional thin-shell result. A finite-thickness transition is the natural setting in which the obstruction could change or dissolve. A resolved layer may introduce anisotropic stress, spin transport, additional fields, or a varying embedding. Smoothing alone does not guarantee success: the integrated stress and conservation laws must still follow from a declared action.

The volume-versus-areal warning from the Introduction follows directly from the independent KS pressure equations [10,11]. In addition to the longitudinal equation displayed above, the angular equation is

$$\kappa p = -\left(\frac{dH_A}{dt_C} + \frac{dH_B}{dt_C} + H_A^2 + H_A H_B + H_B^2\right).$$

Subtracting the two pressure equations and defining $s = H_A - H_B$ gives

$$\frac{ds}{dt_C} + \Theta s = \frac{1}{B^2}. \tag{54}$$

The symmetry-sphere curvature sources anisotropy, so $\Theta > 0$ does not imply $H_B > 0$. This kinematic implication is not part of Theorem 1, but it prevents the junction calculation from being reinterpreted as evidence for a bounce or child universe.

10. Conclusions

We proved a strict surface-energy obstruction for the declared class of comoving and moving timelike boundaries between a homogeneous Kantowski–Sachs region and the ordinary asymptotically flat Schwarzschild exterior. The KS product geometry places its symmetry spheres in the non-untrapped sector throughout the region. For finite rapidity and $F > 0$, the child angular curvature is strictly smaller in magnitude than the parent contribution. Retaining the region containing spatial infinity fixes the parent sign, making the angular jump positive for either child-side orientation.

Negative Israel surface density, $\sigma < 0$, is therefore universal under the theorem's ordinary retained exterior, $F > 0$, finite-rapidity, timelike-embedding, and declared-orientation hypotheses. WEC and DEC fail independently of the surface pressure. NEC violation is not universal: it requires the additional trajectory condition $\Delta K_\tau - \Delta K_\theta < 0$. Proposition 2 proves that for the exact static comoving family this condition holds precisely outside the Schwarzschild photon sphere, $R_0 > 3m$, with saturation at $R_0 = 3m$; the $2+1$-dimensional SEC has the same threshold within that family. Its Nariai-type bulk witness has $\Lambda = R_0^{-2}$, a scale specific to that auxiliary realization and absent from Theorem 1. The surface conservation identity closes, the turning-point limit preserves the obstruction, and the exterior curvature agrees in Schwarzschild and ingoing Eddington–Finkelstein coordinates.

This is neither a new general spherical-junction formalism nor a universal no-go theorem for black-hole-to-cosmology transitions. It leaves open interior and spacelike junctions, alternative global gluings, and finite-thickness layers. Its value is narrower: it extracts from established machinery a strict KS/Schwarzschild surface-energy result and closes the orientation, conservation, turning-point, and chart-regularity checks needed to use it without hidden branch assumptions.

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Software used in this work includes Python, SymPy, SciPy, ReportLab, PGFPlots, and XeLaTeX.

Data and code availability

No observational or experimental datasets were generated or analyzed in this study. The symbolic derivations, numerical checks, figure sources, and one-command verification suite supporting the results are available from the corresponding author upon reasonable request.

A. Orthonormal-frame check

In the child t_C-χ plane, use

$$u^{\hat{a}} = (\gamma, X), \qquad n^{\hat{a}} = \epsilon_C(X, \gamma). \tag{A1}$$

Writing $X = \sinh\eta$ and $\gamma = \cosh\eta$ gives $\dot{\eta} = \dot{X}/\gamma$. Normal projection of the shell acceleration yields

$$n_{\hat{a}}a^{\hat{a}} = \epsilon_C\left(\frac{\dot{X}}{\gamma} + H_A X\right), \tag{A2}$$

which reproduces the temporal component in Eq. (17). Since B depends only on t_C,

$$K^{\theta}{}_{\theta,C} = \frac{1}{B}n^{\mu}\nabla_{\mu}B = \epsilon_C X H_B, \tag{A3}$$

reproducing the angular component and providing a derivation independent of the coordinate calculation.

B. Israel component algebra

Spherical symmetry gives

$$[K^a{}_b] = \mathrm{diag}(\Delta K_\tau, \Delta K_\theta, \Delta K_\theta), \qquad [K] = \Delta K_\tau + 2\Delta K_\theta. \tag{B1}$$

The mixed τ-component of Eq. (8) gives

$$-2\Delta K_\theta = \frac{8\pi G}{c^4}\sigma, \tag{B2}$$

while either angular component gives

$$-\Delta K_\tau - \Delta K_\theta = -\frac{8\pi G}{c^4} p_s. \tag{B3}$$

These are Eq. (37). Substitution into all three mixed equations gives zero residual.

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